Mattia Martini

Postdoctoral researcher

Contact info

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Experience

Postdoctoral researcher Sept 2025 – now

Centre de Mathématiques Appliquées (CMAP), École polytechnique

- Mentors: Mao Fabrice Djete and Huyên Pham
- Part of the ANR PEPR project MIRTE

Postdoctoral researcher Apr 2023 - Aug 2025

Laboratoire J.A. Dieudonné (LJAD), Université Côte d'Azur

- Mentor: François Delarue
- Member of the ERC project ELISA

Education

PhD **Mathematical sciences**, Università degli studi di Milano — Oct 2019 – Mar 2023

- Thesis: *Advances in stochastic analysis on spaces of measures: Kolmogorov equations related to stochastic filtering and mean field optimal stopping*
- Supervisor: Marco Fuhrman

MSc **Mathematical engineering**, Politecnico di Milano (110/110 cum laude) — Sept 2017 – Oct 2019

- Thesis: *Kolmogorov Equations on Space of Measures Associated to Stochastic Filtering Processes*
- Supervisors: Sandro Salsa and Marco Fuhrman

BSc **Mathematical engineering**, Politecnico di Milano (110/110) — Sept 2014 – Sept 2017

- Thesis: *Quantum bits and superdense coding*
- Supervisor: Matteo Gregoratti

High school diploma, I.T.I.S Galileo Galilei, (100/100 cum laude) — 2009 – 2014

Publications

Preprints

[1] N. Fontana, M.M., and H. Pham. “Linear-quadratic mean field games with Volterra dynamics”. arXiv:2610.37622. 2026.

[2] M. F. Djete and M. M. “W-BSDEs with Common Noise: Well-posedness, Characterization, and Consequences”. arXiv:2610.06593. 2026.

[3] M. F. Djete and M. M. “A McKean–Vlasov semigroup and its application to a martingale representation problem”. arXiv:2608.03412. 2026.

[4] F. Delarue, M. M., and G. E. Sodini. “HJB equations driven by the Dirichlet-Ferguson Laplacian in Wasserstein-Sobolev spaces”. arXiv:2511.03522. 2025.

[5] A. Cosso, M. M., and L. Perelli. “Mean convergence rates for Gaussian-smoothed Wasserstein distances and classical Wasserstein distances”. arXiv:2504.17477. 2025.

Published papers

[6] A. Cecchin, S. Daudin, J. Jackson, and M. M. “Quantitative convergence for mean field control with common noise and degenerate idiosyncratic noise”. *Electron. J. Probab.* 30 (2025), pp. 1–62.

[7] G. Franzese, M. M., G. Corallo, P. Papotti, and P. Michiardi. “Latent Abstractions in Generative Diffusion Models”. *Entropy* 27.4 (2025).

[8] F. Delarue and M. M. “Fourier Galerkin approximation of mean field control problems”. *Nonlinear Differential Equations and Applications NoDEA* 32.1 (2024), p. 11.

[9] M. M. “Kolmogorov equations on the space of probability measures associated to the nonlinear filtering equation: the viscosity approach”. *Stochastic Analysis and Applications* 42.6 (2024), pp. 987–999.

[10] A. Cosso and M. M. “On smooth approximations in the Wasserstein space”. *Electronic Communications in Probability* 28 (2023), pp. 1–11.

[11] B. Djehiche and M. M. “Time-inconsistent mean-field optimal stopping: A limit approach”. *Journal of Mathematical Analysis and Applications* 528.1 (2023).

[12] M. M. “Kolmogorov equations on spaces of measures associated to nonlinear filtering processes”. *Stochastic Processes and their Applications* 161 (2023), pp. 385–423.

[13] M. M. and G. E. Sodini. “Numerical Methods for a System of Coupled Cahn-Hilliard Equations”. *Commun. Appl. Ind. Math.* 12 (2021), pp. 1–12.

Teaching

Stochastic Calculus and Mathematical Finance, Universita degli Studi di Torino 2026
Lecturer (25h) | MSc Quantitative Finance and Insurance

Stochastic differential equations, Politecnico di Milano 2022
Teaching assistant (30h) | MSc Mathematical Engineering

Stochastic differential equations, Politecnico di Milano 2021
Teaching assistant (30h) | MSc Mathematical Engineering

Basics of Statistics, Politecnico di Milano 2021
Teaching assistant (30h) | BSc Biomedical Engineering

Stochastic differential equations, Politecnico di Milano 2020
Teaching assistant (30h) | MSc Mathematical Engineering

Probability and Statistics, Politecnico di Milano 2020
Teaching assistant (22h) | BSc Management Engineering

Talks and Presentations

Conferences and Workshops

PGMO Days, EDF Lab Paris-Saclay — Nov 2026
In the invited session "Recent Advances in Mean Field Games and Mean Field Control"

Fifth Italian Meeting on Probability and Mathematical Statistics, Università di Palermo — June 2026
In the session "Stochastic Systems, Mean-Field Models, and Control"

WS Mean-field systems in finance, neurosciences and AI, Folgarida, Trento — Jan 2026
Contributed talk

Interacting Particle Systems and Applications, Università di Siena — Sept 2025
Invited talk

16th Viennese Conference Optimal Control and Dynamic Games, TU Wien — July 2025
In the invited session "Stochastic Control and Applications"

Durham Symposium on Mean Field Games, Durham University — March 2025
Invited talk

WS Optimal Transport: from robust pricing to model calibration, Folgarida, Trento — Feb 2025
Contributed talk

2nd AMS-UMI International Joint Meeting , Università degli Studi di Palermo — July 2024
In the invited session "Mean Field Games and related topics"

Fourth Italian Meeting on Probability and Mathematical Statistics, Roma — June 2024
In the invited session "Stochastic optimal control of McKean-Vlasov equations"

Young Researchers in Stochastic Control and Games , Universität Bielefeld — Dec 2023
Invited talk

PGMO Days, EDF Lab Paris-Saclay — Nov 2023
In the invited session "Mean field games and control"

A Random Walk in the Land of Stochastic Analysis and Numerical Probability, CIRM Luminy — Sept 2023
Poster

SPDEs, optimal control and mean field games , Universität Bielefeld — July 2023
Poster

The 9th International Colloquium on BSDEs and Mean Field Systems, IAE Savoie Mont Blanc — July 2022
Contributed talk

Third Italian Meeting on Probability and Mathematical Statistics, Università di Bologna — June 2022
In the session "New trends in stochastic control"

100 years UMI and First UMI meeting of PhD students , Università di Padova — May 2022
Contributed talk

Trento-Wuppertal School on Stochastic Processes, Analysis and Semigroups, Università di Trento — Sept 2021
Contributed talk

Local Seminars

Séminaire de l'équipe EDP, Université Savoie Mont Blanc (LAMA)) — Nov 2026
Seminar of the PDE group

Demi-journée MIRTE, Institut Louis Bachelier — Sept 2026

Seminar of the ANR PEPR MIRTE

Torino seminar series in Stochastics and Mathematical Statistics, Università di Torino — Apr 2026
Seminar of the probability and statistics group

Seminario di Probabilità e Statistica, Politecnico di Milano — Jan 2026
Seminar of the probability and statistics group

Seminari in Probabilità e Finanza, Università di Padova — Dec 2025
Seminar of the probability and finance group

Séminaire du Pôle Probabilités, École polytechnique (CMAP) — Oct 2025
Seminar of the probability group

Probability and Statistics Seminar, University of Luxembourg — Jan 2025
Seminar of the probability and statistics group

PDE Seminar, Georgia Tech (online) — Sept 2024
Seminar of the PDE group

Seminario di Probabilità e Statistica, Politecnico di Milano — June 2024
Seminar of the probability and statistics group

Séminaire de Probabilités et Statistique, Université Côte d'Azur (LJAD) — Dec 2023
Seminar of the probability and statistics group

Seminars cycle "Insalate di Matematica", Università degli Studi Milano-Bicocca — Nov 2021
Seminar organized by the PhD students

Visiting periods

Università degli Studi di Torino, 1 Month — Apr-May 2026
Visiting professor position, ESOMAS department

University of Luxembourg, 1 Week — Jan 2025
Collaboration with Michele Stecconi

Imperial College, London, 1 Week — Jul 2024
Collaboration with Ofelia Bonesini

Università di Parma, Italy, 1 Week — Feb 2024
Collaboration with Alessandro Calvia

KTH, Stockholm, 1 Week — May 2022
Collaboration with Boualem Djehiche

Organization

Contributed session "(Stochastic) Analysis on Spaces of Probability Measures and Applications", with G. E. Sodini
5th Italian Meeting on Probability and Mathematical Statistics, 2026 | Palermo

Workshop on Stochastic Processes, Stochastic Optimal Control, and their Applications, among the organizers
September 26 - 27th, 2024 | Politecnico di Milano

Contributed session "Mean Field Games and Mean Field Control I", with J. Dianetti
4th Italian Meeting on Probability and Mathematical Statistics, 2024 | Roma

Phd Days 2021, among the organizers
December 14 - 17th, 2021 | Università degli Studi di Milano

Others

Abilitations: Maître de conférences (26 - Mathématiques appliquées et applications des mathématiques)

Referee activities:

- Annals of Applied Probability, Applied Mathematics and Optimization, SICON, Stochastic Processes and Their Applications, Journal of Differential Equations, Systems & Control Letters, Statistics & Probability Letters
- ANR (French grant)

Participation to research projects:

- Sept 2025 - now: part of the ANR PEPR project MIRTE (ANR-23-EXMA-0011)
- Apr 2023 - Aug 2025: member of the ERC ADV project ELISA (Grant agreement No. 101054746), P.I. F. Delarue.

Memberships:

- From 2025: Gruppo UMI Prisma (PRobability In Statistics, Mathematics and Applications)
- From 2022: Unione Matematica Italiana (UMI)
- From 2020: Indam, GNAMPA

Languages: Italian (Native) | English (Full professional proficiency) | French (Intermediate)

October 28, 2026
Mattia Martini